Bassez 2021 Breast Cancer — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

41 samples intermixed on one continuous manifold; sample iLISI 6.1→9.1 (of 41) after Harmony θ=10; 363 island cells removed; cycling orthogonal to modules.

Gate 2 — module geography → GO (non-branching)

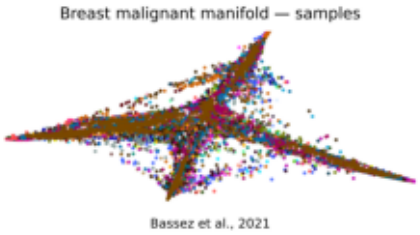
Single collapsed axis: pSom/Gast developmental origin → Crest terminal (pSom↔Crest r=−0.82). No Gast arm at any tip; PPT tree over-branches — not a bifurcation.

Gate 3 — hub cross-repression → NA

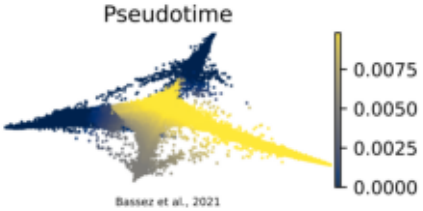
Non-branching geometry — no Gast/pSom competing arms to test.

STEP FIGURES

Manifold — samples



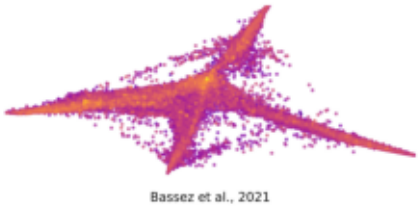
Pseudotime



Gastrulation



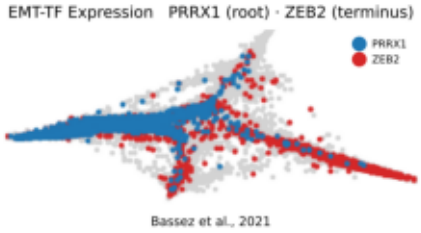
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Bassez et al., 2021), 10x — 41 samples, 65,519 malignant cells.

Verdict.
Pattern reproduces as the non-branching case — one embryonic axis, pSom/Gast origin → Crest terminus. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw): Gast −0.014 / pSom +0.022 / Crest −0.005. EMT-TFs PRRX1 (early/root) · ZEB2 (late/terminus).

Parameter choices

Sample threshold	16 (gap-finding) → 41/42 kept	**Embedding**	overdispersed 2498; n_pcs 20; Harmony θ=10 on sample; n_neighbors 30; UMAP spread 1.3; islands 363 removed
Trajectory	scFates curve on diffusion rep, Nodes=15, λ=0.3, μ=100; fold-free (self-approach 2.56)	**Root**	pSom/Gast-high tip (pt 0.28); Crest terminal (0.74)
Annotation	Bassez2021 malignant meta-programs (EMT I dominant) + data-derived compartments	**Module means**	Gast −0.014 / pSom +0.022 / Crest −0.005 (raw)
EMT-TFs	PRRX1 (root) · ZEB2 (terminus)		